

VISVESVARAYA TECHNOLOGICAL UNIVERSITY

"Jnana Sangama", Belgaum - 590014, Karnataka



MOBILE APPLICATION DEVELOPMENT REPORT

on

INTELLICOURSE: A FULL-STACK LEARNING MANAGEMENT SYSTEM MOBILE APPLICATION

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in partial fulfillment for the award of the degree of

BACHELOR OF ENGINEERING

in

COMPUTER SCIENCE AND ENGINEERING



B.M.S. COLLEGE OF ENGINEERING

(Autonomous Institution under VTU)

BENGALURU - 560019

Feb-2026 to June-2026

B. M. S. College of Engineering

Bull Temple Road, Bangalore 560019

(Affiliated to Visvesvaraya Technological University, Belgaum)



Department of Computer Science and Engineering

CERTIFICATE

This is to certify that the project work entitled "IntelliCourse: A Full-Stack Learning Management System Mobile Application" carried out by Harshith M (1BM24CS113), Harshith D (1BM24CS112) and Hanish S Adhi (1BM24CS109), who are bonafide students of B. M. S. College of Engineering, is submitted in partial fulfillment for the award of Bachelor of Engineering in Computer Science and Engineering during the academic semester Feb-2026 to June-2026.

Signature of the Guide

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B.M.S. COLLEGE OF ENGINEERING
DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

DECLARATION

We, Harshith M (1BM24CS113), Harshith D (1BM24CS112) and Hanish S Adhi (1BM24CS109), students of 4th Semester, B.E., Department of Computer Science and Engineering, B. M. S. College of Engineering, Bangalore, hereby declare that this Mobile Application Development report entitled "IntelliCourse: A Full-Stack Learning Management System Mobile Application" has been carried out by us under the guidance of Dr. Kavitha Sooda.

We also declare that, to the best of our knowledge and belief, the work reported here is original and has not been submitted as part of any other report by any other students.

Signature

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Abstract

IntelliCourse is a full-stack Learning Management System designed to support online course discovery, enrollment, lecture delivery, instructor course management, secure media upload, payment processing and learner progress tracking. The system contains a Node.js and Express backend with MongoDB persistence, JWT-based authentication, role-based access control for students, instructors and administrators, and security middleware for CORS, rate limiting, HTTP headers, XSS mitigation and request sanitization. The mobile application is built using Expo, React Native and TypeScript with Expo Router navigation and Zustand state management. Students can browse published courses, enroll or purchase access, view lessons, stream videos through secure playback URLs and monitor completion percentage. Instructors can create courses, add lectures, upload thumbnails and videos through presigned S3 URLs, and manage their teaching catalogue. A React web client is also included for desktop course browsing, authentication, instructor dashboards and learning views. The project demonstrates a practical mobile-first LMS architecture with reusable components, well-defined REST APIs, payment workflow integration and database models for users, courses, lectures, purchases, media assets and progress records.

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Chapter 1

Introduction

IntelliCourse is an online learning platform that combines a mobile application, web client and secure API server to provide a complete LMS experience. The application helps learners discover courses, enroll in structured lessons, watch video content and track course completion. It also gives instructors a controlled workspace for creating and maintaining courses, adding lecture content and monitoring teaching activity.

1.1 Problem Statement

Many learning platforms separate course browsing, video delivery, payment status and progress tracking into disconnected workflows. IntelliCourse addresses this by integrating authentication, course management, payment, media upload and learner progress into one mobile-first platform.

1.2 Users Supported

- Student: browses the catalog, purchases or enrolls in courses, watches lessons and tracks completion.
- Instructor: creates courses, manages lectures, uploads media and monitors course activity.
- Administrator: has elevated access for protected course-management and system operations.

1.3 Student Functionalities

1. Create an account and sign in using validated credentials.
2. Browse published courses by category and search query.
3. View course detail pages with instructor, price, level and lecture information.
4. Enroll in free courses or complete the payment flow for paid courses.
5. Stream lecture videos through authenticated playback URLs.
6. Mark lessons complete and view course progress percentage.
7. Edit profile information and avatar.

1.4 Instructor Functionalities

8. Create new courses with title, subtitle, description, category, level and price.
9. Upload thumbnails and lecture videos through presigned AWS S3 URLs.
10. Add, list and delete lectures in owned courses.
11. View instructor dashboard metrics and manage created courses.
12. Update course metadata and control course availability.

1.5 System Architecture

The architecture connects the Expo mobile client and React web client to a secured Express API. MongoDB stores application data, AWS S3 stores media files, and Razorpay/Stripe workflows handle purchases.

IntelliCourse System Architecture

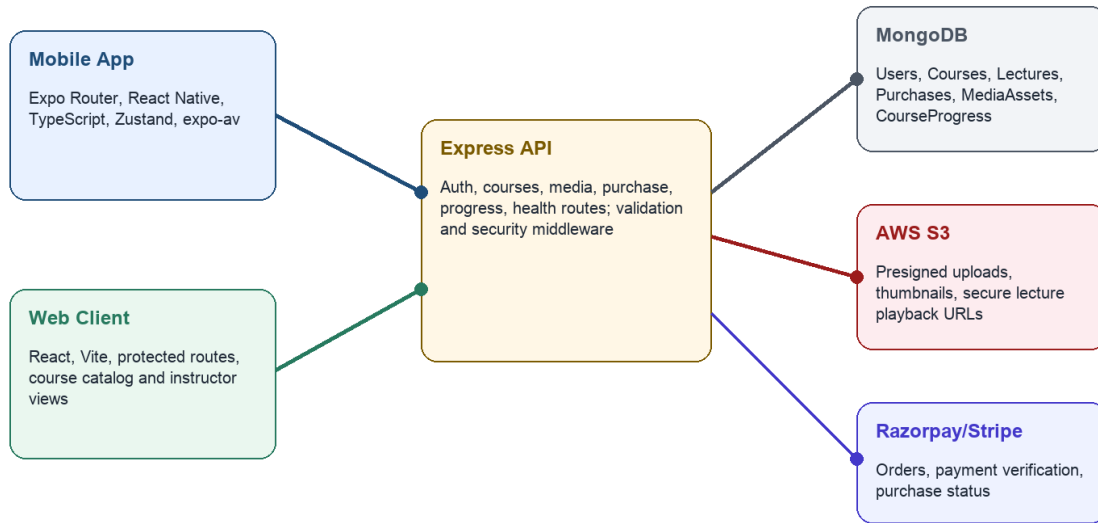


Fig 1.1 System architecture of IntelliCourse

Chapter 2

Hardware and Software Requirements

2.1 Hardware Requirements

Component	Minimum Requirement	Purpose
Development Machine	Intel i5 / Apple Silicon equivalent, 8 GB RAM	Run backend, web client, mobile simulator and build tools.
Mobile Device / Emulator	Android or iOS device with network access	Test Expo mobile app, video playback and profile/media flows.
Server Environment	Cloud VM or managed Node.js hosting	Host Express API and connect to MongoDB, AWS and payment providers.
Network	Stable broadband connection	Upload video media, fetch presigned URLs and test payment integrations.

2.2 Software Requirements

Layer	Technology	Description
Backend	Node.js, Express, Mongoose	REST API server, route handling, MongoDB data modelling and middleware.
Database	MongoDB	Stores users, courses, lectures, purchases, media assets and progress records.
Mobile	Expo, React Native, TypeScript	Cross-platform mobile screens, navigation, forms and video playback.
State/API	Zustand, Axios	Client-side state and backend communication with token injection.
Web Client	React, Vite, React Router	Desktop interface for catalog, authentication, dashboards and course learning.
Security	JWT, bcryptjs, Helmet, CORS, rate-limit	Authentication, password hashing and API hardening.
Media	AWS S3 SDK, Multer, Expo Image Picker	Thumbnail/video upload, presigned URLs and profile image selection.
Payments	Razorpay, Stripe	Course checkout and verification.

Chapter 3

Design Layouts: Screen Shots of Mobile App / Webpages

3.1 Login Screen

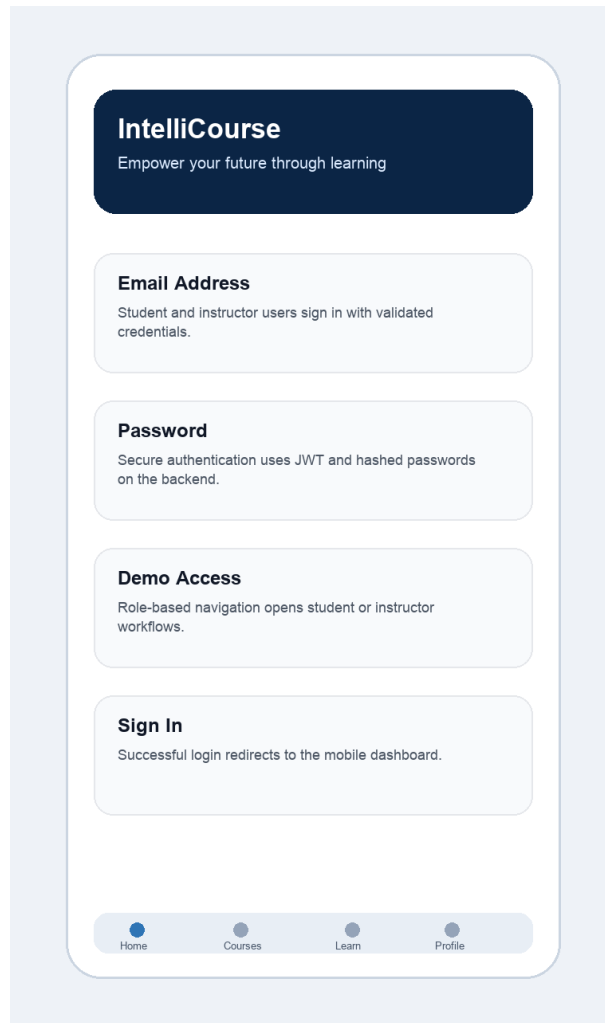


Fig 3.1 Login screen for student and instructor authentication

3.2 Student Home Screen

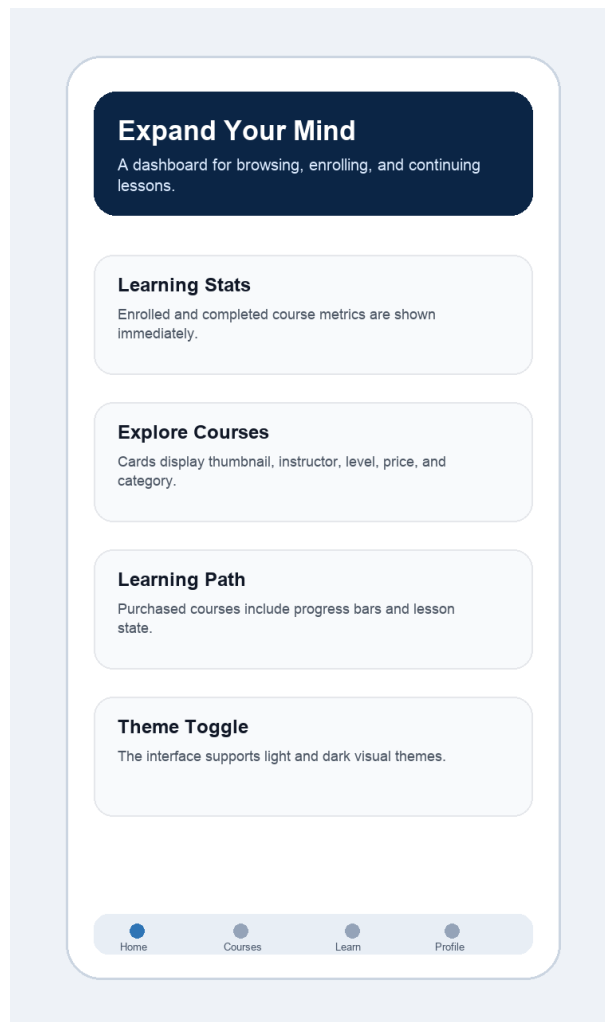


Fig 3.2 Student dashboard and course discovery screen

3.3 Instructor Management Screen

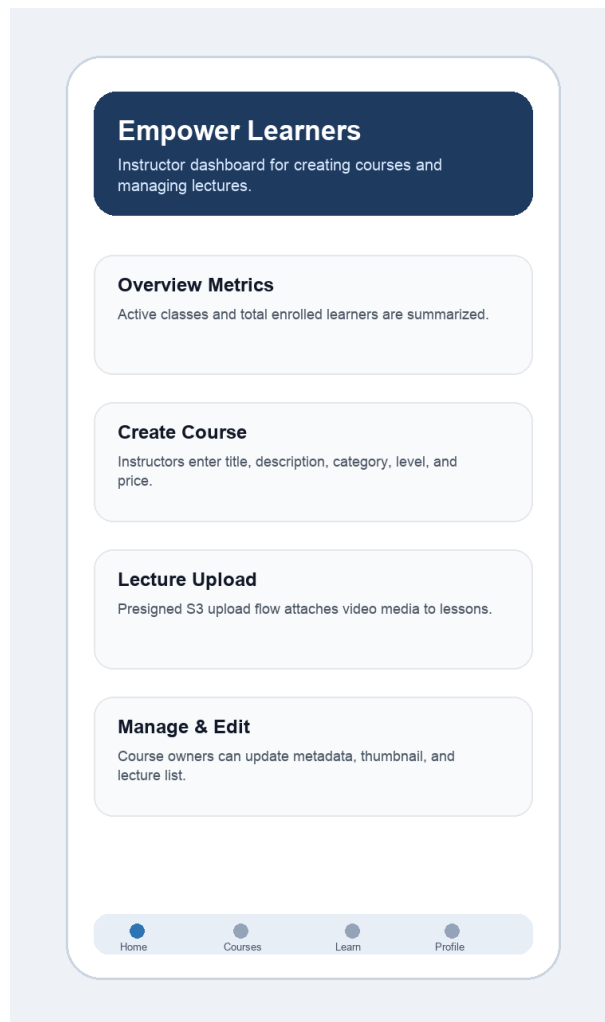


Fig 3.3 Instructor course creation and management screen

3.4 Web Client Pages

- Landing page: displays IntelliCourse branding, featured courses and category tracks.
- Courses page: supports catalog search, category filtering and course card display.
- Course detail page: displays instructor, price, lectures and purchase/access action.
- Instructor dashboard and course editor: support protected course management workflows.

Chapter 4

Database Table Screen Shots

The project uses MongoDB collections defined through Mongoose schemas. The following tables document the main collections and their fields.

4.1 User Collection: Description

Field	Type	Description
name	String	Required student/instructor name.
email	String	Unique login identifier with email validation.
password	String	Hashed password stored with select:false.
role	Enum	student, instructor or admin.
enrolledCourses	Array	Course references and enrollment date.
createdCourses	Array	Course references created by instructor.

4.2 Course Collection: Description

Field	Type	Description
title	String	Course title with max length validation.
category	String	Required catalog category.
level	Enum	beginner, intermediate or advanced.
price	Number	Non-negative course price.
instructor	ObjectId	Reference to User.
lectures	Array	References to Lecture documents.
isPublished	Boolean	Controls public visibility.

4.3 Lecture Collection: Description

Field	Type	Description
title	String	Lecture title.

videoUrl	String	Stored media playback/source URL.
duration	Number	Rounded lecture duration.
s3Key	String	AWS S3 object key.
mediaAsset	ObjectId	Reference to MediaAsset.
order	Number	Required lecture sequence order.

4.4 CourseProgress Collection: Description

Field	Type	Description
user	ObjectId	Learner reference.
course	ObjectId	Course reference.
completionPercentage	Number	Auto-calculated percentage between 0 and 100.
lectureProgress	Array	Per-lecture completion, watch time and timestamps.
isCompleted	Boolean	True when progress reaches 100 percent.

4.5 ER Diagram

Database ER Diagram

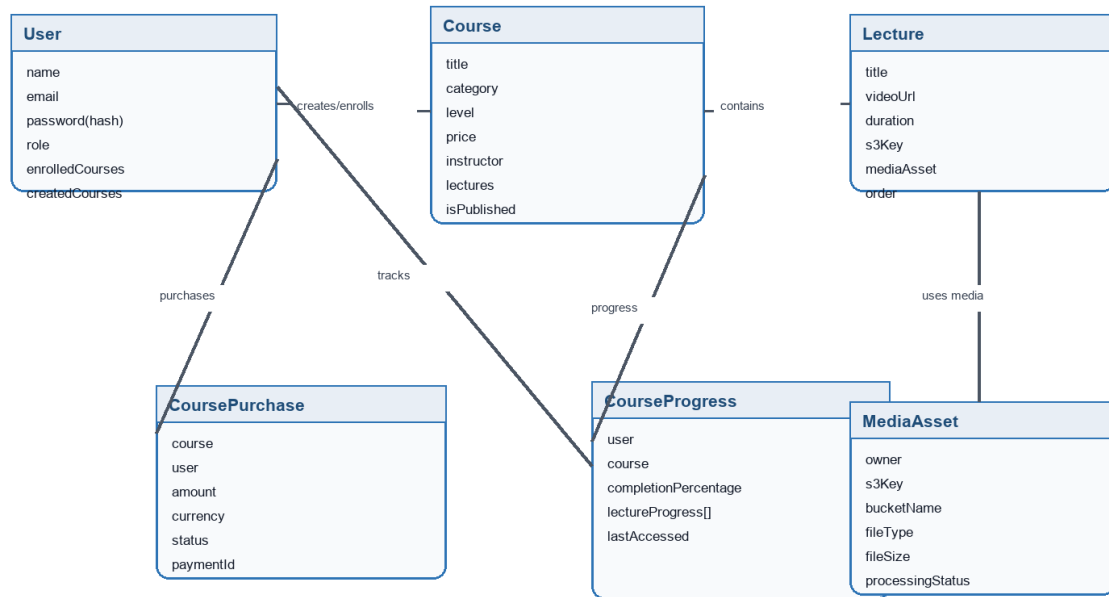


Fig 4.1 ER diagram of IntelliCourse database

Chapter 5

Testing

Testing focuses on authentication, course management, payment/enrollment, media upload and progress tracking. Backend model/controller checks are supported by the repository test file `backend/tests/api.test.js`.

TC ID	Module	Test Scenario	Input Data	Expected Result	Status
TC_001	Auth	Valid login	Registered email and password	Token issued; dashboard opens.	Pass
TC_002	Auth	Weak signup password	Password under 8 chars	Validation error displayed.	Pass
TC_003	RBAC	Student opens create course	Student account	Access denied or redirected.	Pass
TC_004	Courses	Instructor creates course	Title, category, level, price	Course appears in Manage Courses.	Pass
TC_005	Catalog	Browse published courses	Published courses API	Course cards are returned.	Pass
TC_006	Media	Request upload URL	Course ID and video metadata	Presigned S3 URL returned.	Pass
TC_007	Payments	Enroll/purchase course	Course ID	Enrollment status becomes true.	Pass
TC_008	Progress	Mark lecture complete	Course and lecture ID	Percentage recalculated.	Pass
TC_009	Profile	Update profile	Name, email and avatar	Profile saved successfully.	Pass
TC_010	Security	Invalid token request	Expired JWT	Authentication error returned.	Pass

Chapter 6

Conclusion and Future Work

6.1 Conclusion

IntelliCourse successfully demonstrates a mobile-first LMS with a secure backend, role-based workflows, course creation, lecture management, payment support and progress tracking. The system separates concerns cleanly across an Express API, MongoDB data layer, Expo mobile client and React web client. The implementation also includes reusable UI components, typed mobile models, media upload support through AWS S3 and purchase flows through Razorpay/Stripe integrations.

6.2 Future Work

- Add ratings, reviews and recommendation logic for personalized course discovery.
- Introduce certificates after course completion with downloadable PDF generation.
- Improve analytics for instructors with learner retention and completion dashboards.
- Add offline lesson metadata caching and resumable downloads where licensing permits.
- Expand automated test coverage with API integration tests and mobile UI tests.
- Deploy backend, web client and database using production CI/CD workflows.

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